Linear Regression

Václav Šmídl

March 12, 2026

Linear regression and OLS

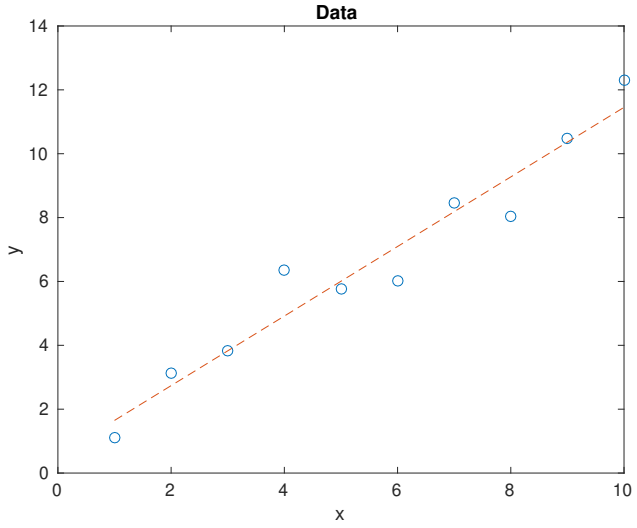
Fit by a linear function:

$$\begin{array}{llll} y_1 & = ax_1 & +b1, & +e_1 \\ y_2 & = ax_2 & +b1 & +e_2, \\ \vdots & \vdots & \vdots & \vdots \end{array}$$

In matrix notation $\theta = [a, b]^T$:

$$\mathbf{y} = X\theta + \mathbf{e},$$

Minimize $\sum_i e_i^2 = \mathbf{e}^T\mathbf{e} = ||\mathbf{y} - X\theta||_2^2$:



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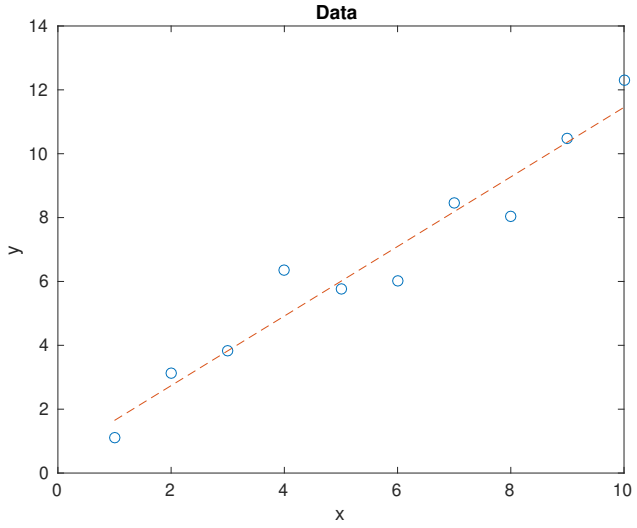

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$$\frac{d}{d\theta}((\mathbf{y} - X\theta)^T(\mathbf{y} - X\theta)) = 0$$

$$\frac{d}{d\theta}(\mathbf{y}^T\mathbf{y} - \theta^T X^T\mathbf{y} - \mathbf{y}^T X\theta + \theta^T X^T X\theta) = 0$$

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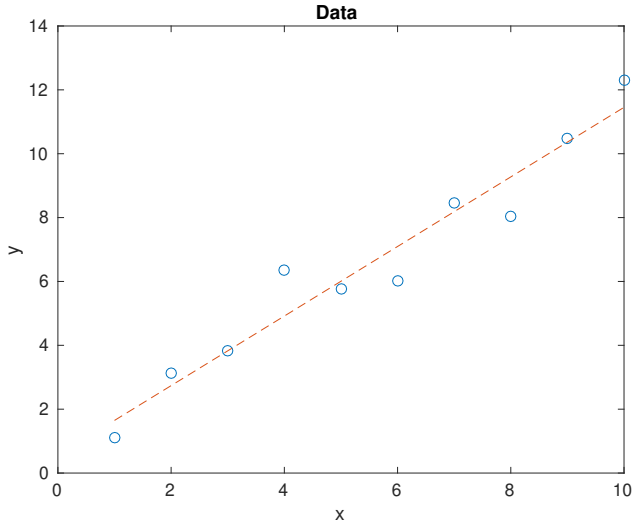


Solution:

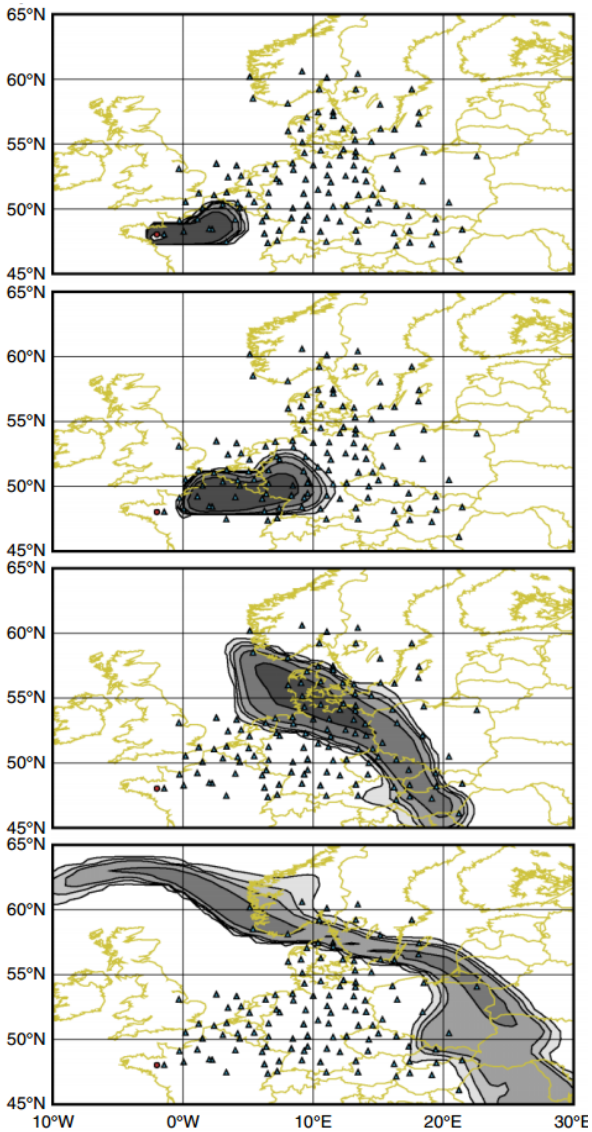
$$\hat{\theta} = (X^T X)^{-1} X^T \mathbf{y}.$$

Example: The European Tracer Experiment

- Release of 340kg of PMCH for 12 hours

The tracer release unit

- **Direct modeling:**
  with known meteo and release rates
- **Inverse:**
  Find release profile, how much and when released



Instantaneous release

Consider release of 1kg of the tracer from a known location and known time instance t_0. We can run an atmospheric transport model to predict concentration at sensors.

- For each sensor location and time $y_{l,\tau}$ we compute prediction $\hat{y}_{l,\tau,t_0}$.
- If we do not know how much tracer was released, we can assume that $2\times$ more tracer would imply $2\times$ greater sensor response. Unknown $\theta_0$kg:

  $$\boldsymbol{y} = \hat{\boldsymbol{y}}_{t_0}\theta_0,$$

  i.e. a least squares problem.
- If the release has happened in three times, t_0, t_1, t_2 the response should be

  $$\boldsymbol{y} = \hat{\boldsymbol{y}}_{t_0}\theta_0 + \hat{\boldsymbol{y}}_{t_1}\theta_1 + \hat{\boldsymbol{y}}_{t_2}\theta_2,$$
- To estimate the time of the release, we define a grid of point releases $[t_1, \dots t_n]$ and assume that the amount of the releases tracer is unknown $[\theta_1 \dots \theta_n]$

  $$\boldsymbol{y} = X\boldsymbol{\theta}, \quad X = [\hat{\boldsymbol{y}}_{t_1}, \hat{\boldsymbol{y}}_{t_2}, \dots, \hat{\boldsymbol{y}}_{t_n}]$$

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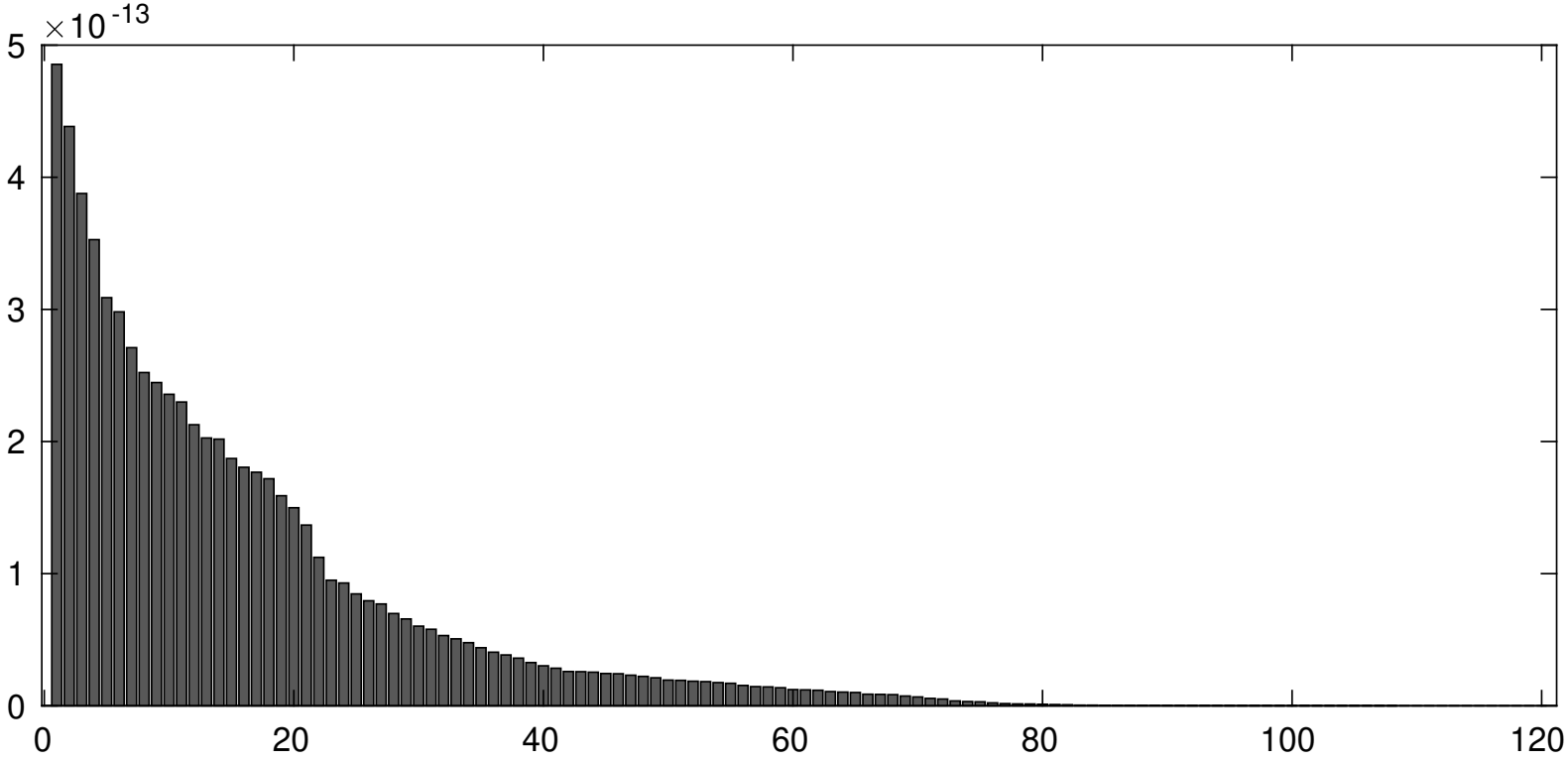

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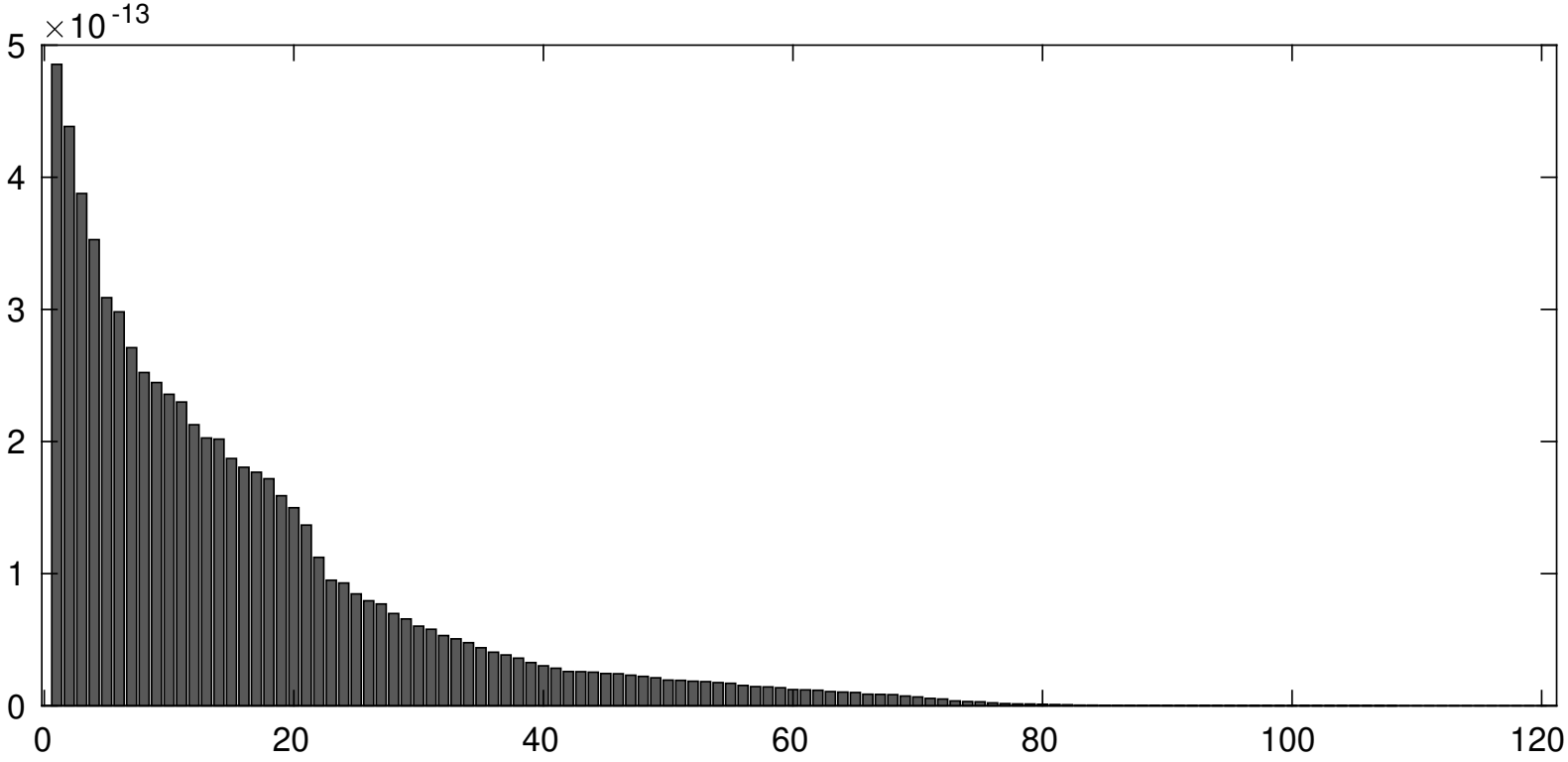

3. Inverse: $\left(X^\top X\right)^{-1} = \left(U^\top \mathrm{diag}(\boldsymbol{s}^2)U\right)^{-1} = U^\top \mathrm{diag}(\boldsymbol{s}^{-2})U$
4. Result is dominated by the smallest singular value (eigenvalue)

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- Solution of optimization problem:

$$\hat{\theta} = \arg\min_{\theta} \left(||\mathbf{y} - X\theta||_2^2 + \alpha ||\theta||_2^2\right),$$

with extreme at:

$$-X^T \mathbf{y} + X^T X \theta + \alpha \theta = 0.$$

Bayesian Ridge regression

Probability model:

$$\begin{aligned} \boldsymbol{y} &= X\theta + \boldsymbol{e}, \quad e_i \sim \mathcal{N}(0, \sigma_e) \\ \boldsymbol{e} &= \boldsymbol{y} - X\theta, \quad p(\boldsymbol{e}) = \mathcal{N}(0, \sigma_e^2 I) \propto \frac{1}{\sigma_e^n} \exp\left(-\frac{1}{2}\frac{\boldsymbol{e}^T\boldsymbol{e}}{\sigma_e^2}\right) \end{aligned}$$

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Independent Gaussian noise

$$p(\boldsymbol{y}|X, \theta, \sigma_e) = \mathcal{N}(X\theta, \sigma_e^2 I) \propto \exp\left(-\frac{1}{2}\sigma_e^{-2}(\boldsymbol{y} - X\theta)^T(\boldsymbol{y} - X\theta)\right)$$

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Using **common** measurement precision $\omega = \frac{1}{\sigma_e^2}$ and prior precision α:

$$\begin{aligned} p(\boldsymbol{y}|X, \theta, \omega) &= \mathcal{N}(X\theta, \omega^{-1} I) \propto \omega^{n/2} \exp\left(-\frac{1}{2}\omega(\boldsymbol{y} - X\theta)^T(\boldsymbol{y} - X\theta)\right) \\ p(\theta|\alpha) &= \mathcal{N}(0, \alpha^{-1} I) \propto \alpha^{p/2} \exp\left(-\frac{1}{2}\alpha\theta^T\theta\right) \end{aligned}$$

where n is the number of measurements and p is the length of θ.

Bayesian Ridge regression

Posterior:

$$
\begin{aligned}
p(\theta|X, y, \alpha) &\propto \exp\left\{-\frac{1}{2}\omega\left(\mathbf{y} - X\theta\right)^T\left(\mathbf{y} - X\theta\right) - \frac{1}{2}\alpha\theta^T\theta\right\} \\
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&= \mathcal{N}(\mu_\theta, \Sigma_\theta) \\
&\quad \propto \exp\left\{-\frac{1}{2}(\theta^T\Sigma_\theta^{-1}\theta - \theta^T\Sigma_\theta^{-1}\mu_\theta - \mu_\theta^T\Sigma_\theta^{-1}\theta + \mu_\theta^T\Sigma_\theta^{-1}\mu_\theta)\right\}
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$$

Posterior

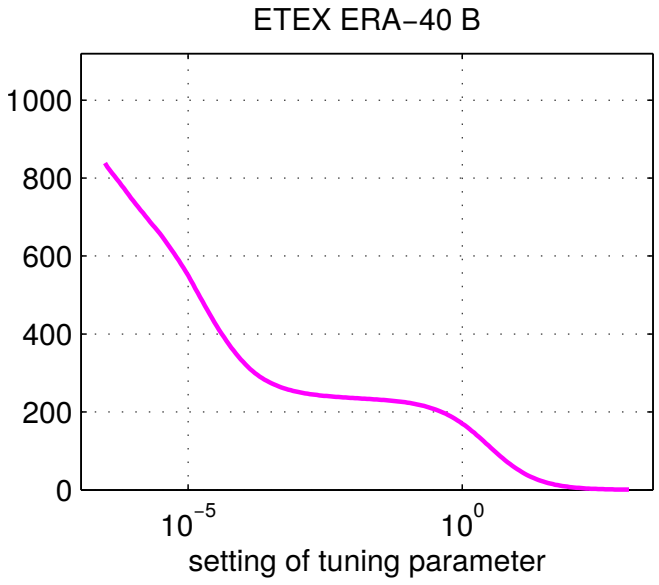
$$
p(\theta|X,\boldsymbol{y},\alpha)=\mathcal{N}((X^TX+\frac{\alpha}{\omega}I)^{-1}X^T\boldsymbol{y},(\omega X^TX+\alpha I)^{-1})
$$

Mean equal to RidgeLS for $\tilde{\alpha}=\frac{\alpha}{\omega}$. Matrix $(\omega X^TX+\alpha I)^{-1}$ had a new interpretation.

Ridge regression – selection of α

Selection of α:

1. Cross-validation
   (analytical solution for α)
   - split data to train&test
   - can be noisy ($100\times$ for each α)
2. L-curve
   plot of $||\theta||_2^2$ versus $||\mathbf{y} - X\theta||_2^2$
3. Log-scale L-curve
   plot $\log ||\theta||_2^2$ versus $\log ||\mathbf{y} - X\theta||_2^2$
4. Joint estimation of α and θ
   - Bayes



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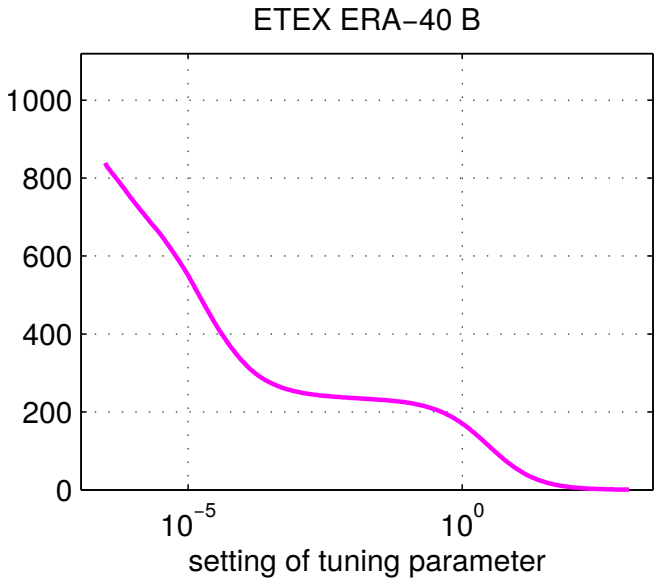

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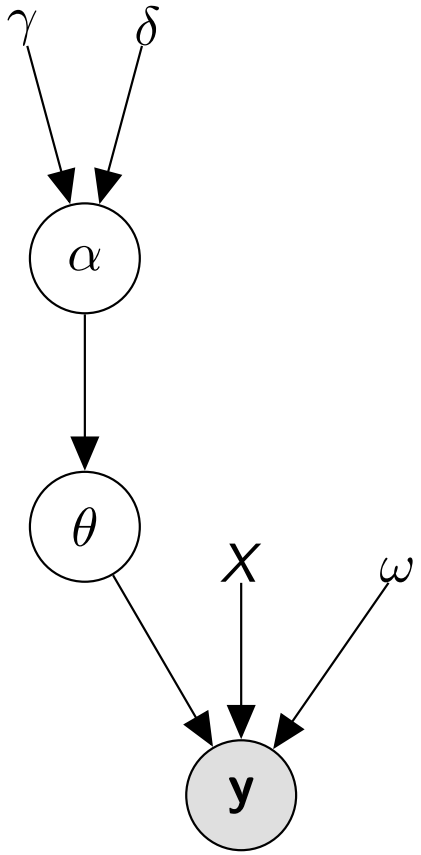

Introduce prior

$$p(\alpha) = G(\delta_0, \gamma_0),$$

compute

$$\hat{\alpha} = \arg\max_{\alpha} p(\alpha | \mathbf{y}, X)$$

or

$$p(\theta | X, \mathbf{y}, \omega) = \int p(\theta, \alpha | X, \mathbf{y}, \omega) d\alpha$$

.

Bayesian estimation of α

Joint likelihood

$$
\begin{aligned}
p(\mathbf{y}, \theta, \alpha | X) &= p(\mathbf{y}|\theta, X, \omega) p(\theta|\alpha) p(\alpha) \\
&= \mathcal{N}(X\theta, \omega I) \mathcal{N}(0, \alpha^{-1} I) G(\delta_0, \gamma_0), \\
p(\theta, \alpha | \mathbf{y}, X) &\propto \alpha^{\delta_0 + \frac{p}{2} - 1} \exp\left\{ -\frac{1}{2}\omega ||\mathbf{y} - X\theta||_2^2 - \frac{1}{2}\alpha ||\theta||_2^2 - \gamma_0 \alpha \right\}
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Conditional for θ:

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p(\theta|\alpha, \mathbf{y}, X) = \mathcal{N}(\hat{\theta}, (X^T X + \alpha I)^{-1}),
$$

Conditional for α:

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p(\alpha|\theta, \mathbf{y}, X) = \mathcal{G}(\delta_0 + \frac{p}{2}, \gamma_0 + ||\theta||_2^2),
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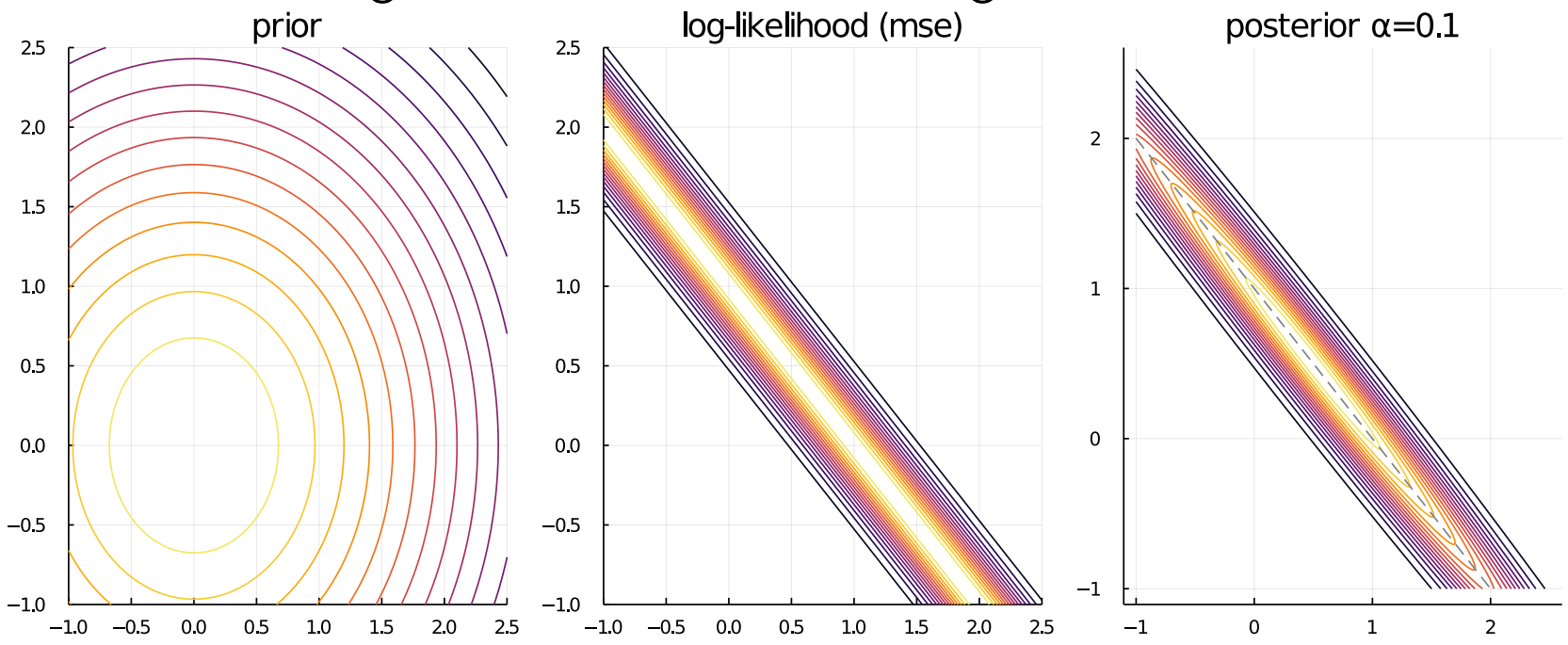
Solution: EM, VB, Gibbs...

Graphical visualization: Ridge

Example:

$$y_1 = x_1\theta_1 + x_1^2\theta_2 \qquad y_1 = 0 \quad x_1 = 0$$
$$y_2 = x_2\theta_1 + x_2^2\theta_2 \qquad y_2 = 1 \quad x_2 = 1$$

Eigenvectors of covariance matrix determine direction of elipsis of posterior HPD, eigenvalues determine the length.



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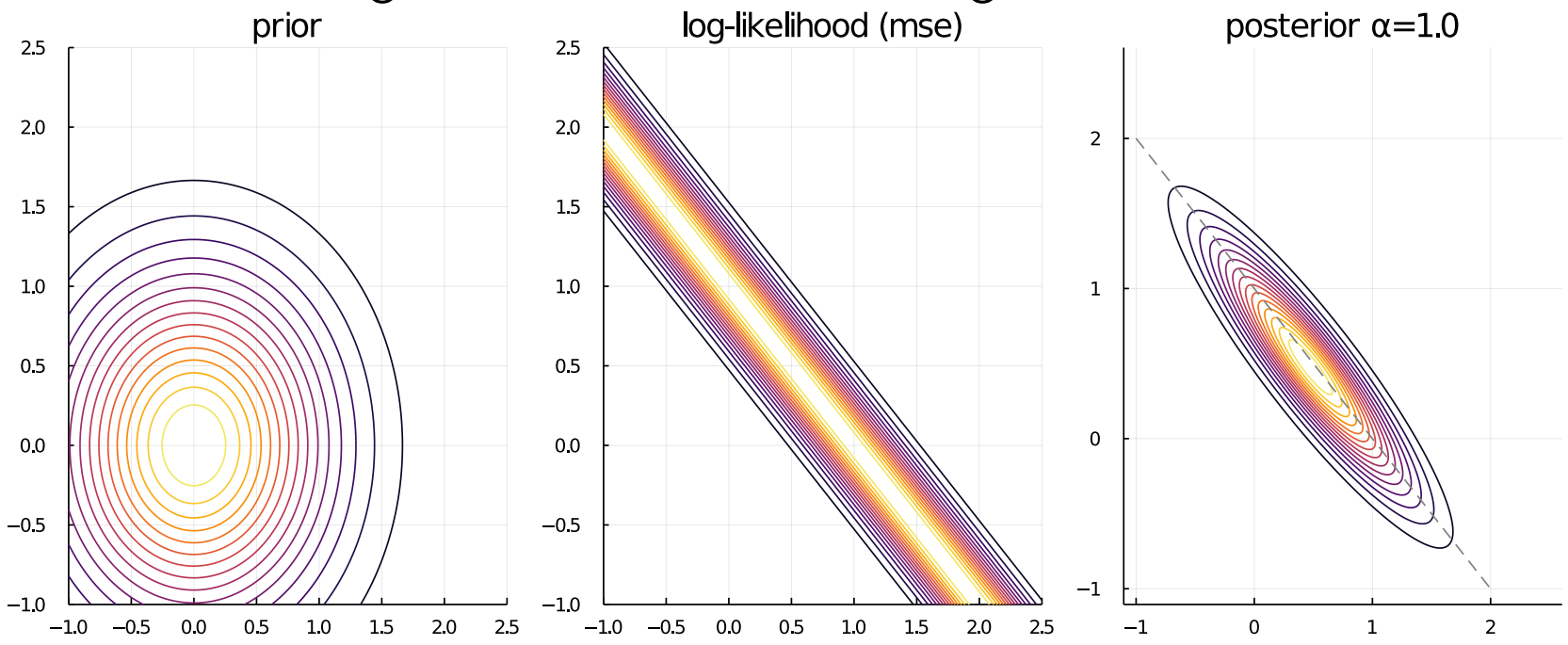

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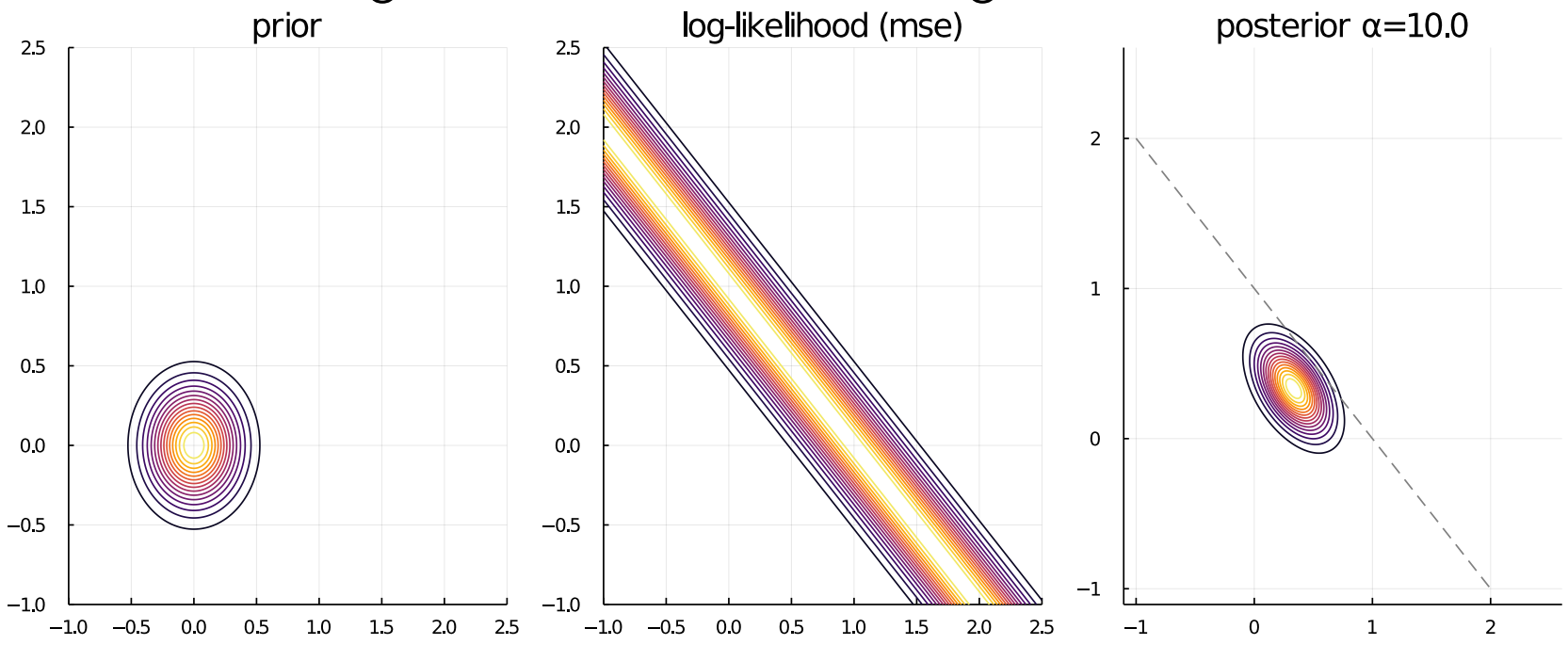

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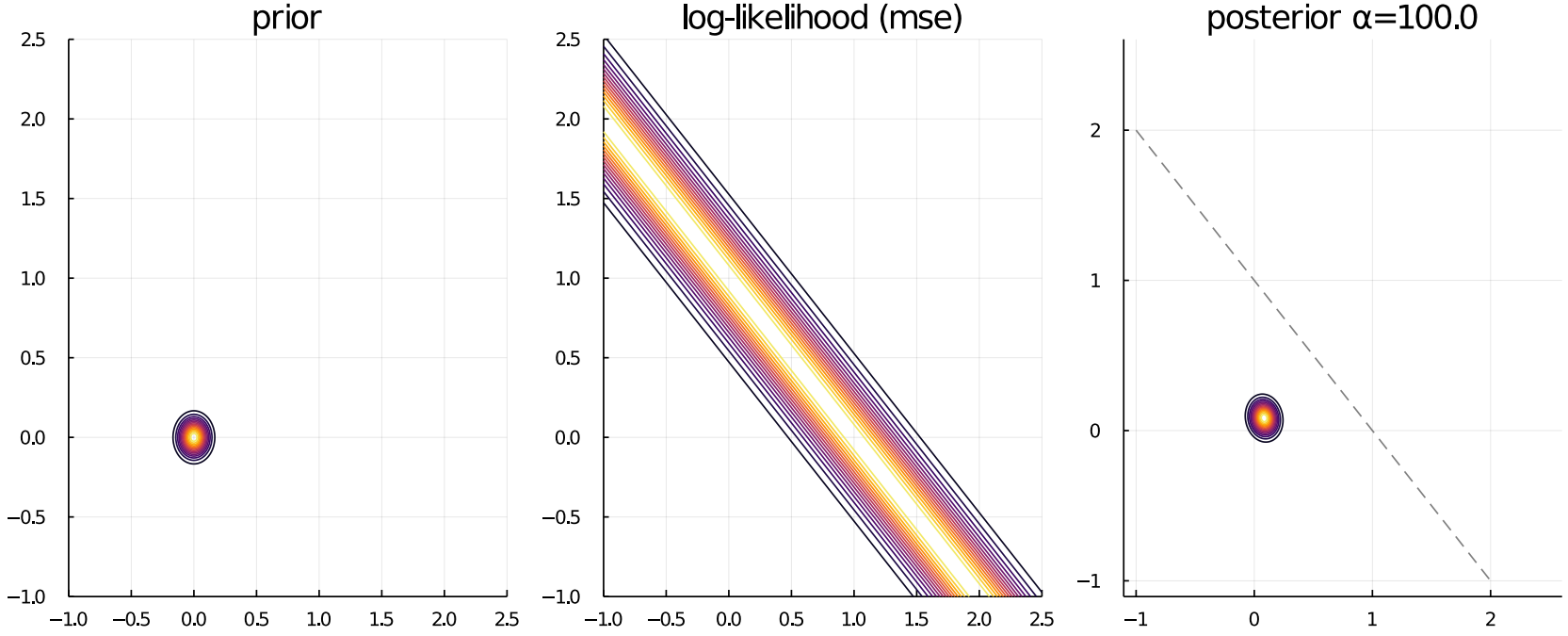

- All solutions lie on line. Solution $y = x$ and $y = x^2$ unlikely.

Encoding different prior information

Gaussian prior favors solutions as close to zero in Eucleidian distance as possible.

- What if we overdo it: $y = x\theta_1 + x^2\theta_2 + x^3\theta_3 + x^4\theta_4 + x^5\theta_5$

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- What if we overdo it: $y = x\theta_1 + x^2\theta_2 + x^3\theta_3 + x^4\theta_4 + x^5\theta_5$
- If we do not know the basis we add as many as possible

We may prefer sparse solutions (with as many zeros as possible)

Spike and slab prior:

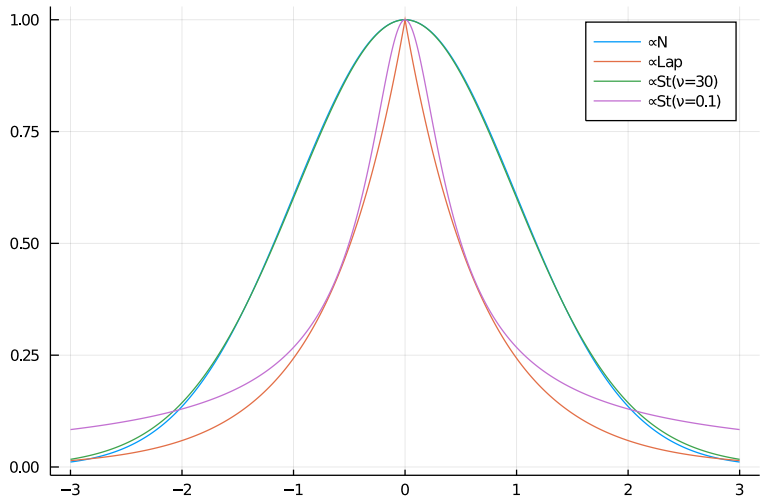
$$p(\theta) = \lambda\mathcal{N}(0, \sigma_0) + (1-\lambda)\mathcal{N}(0, \sigma_1),$$

Laplace prior

$$p(\theta) = (2b)^{-1} \exp\left(-\frac{1}{2b}|x|\right).$$

Student't prior ($\nu \ll 20$):

$$p(\theta) = St(0, \sigma, \nu) = \left(1 + \frac{\theta^2}{\nu\sigma^2}\right)^{-\frac{\nu+1}{2}}.$$



Maximum likelihood approach

Maximum likelihood/aposteriori approach:

$$\hat{\theta}_{ML} = \arg\min_{\theta} \mathcal{L}(\theta) \qquad \mathcal{L}(\theta) = -\log p(\boldsymbol{y}|X, \theta)$$

$$\hat{\theta}_{MAP} = \arg\min_{\theta} \mathcal{L}(\theta) + \mathcal{R}(\theta) \qquad \mathcal{R}(\theta) = -\log p(\theta)$$

Joint likelihood with Laplace prior (LASSO)

$$p(\mathbf{y}, \theta|X, b) \propto \exp\left(-\frac{1}{2}||\mathbf{y} - X\theta||_2^2 - \frac{1}{2b}||\theta||_1\right),$$

Tight connection between optimization-based and Bayesian.

- optimization corresponds to ML/MAP
- Bayesian can be obtained from optimization via energy model

$$p(\theta) \propto \exp(-\mathcal{L}(\theta))$$

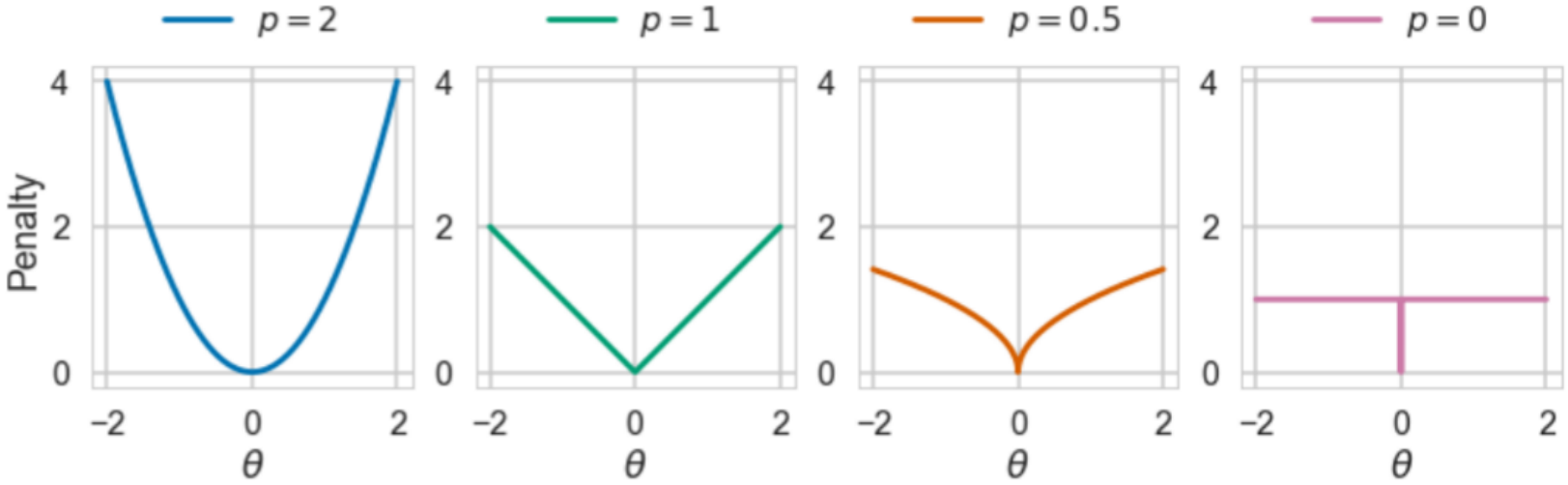
where the challenge is to find the normalization constant.

Ideal sparse regression

What we really want is to penalize non-zeros:

$$\mathcal{R}(\theta) = \begin{cases} 0 & \text{if } \theta = 0, \\ 1 & \text{otherwise} \end{cases}$$

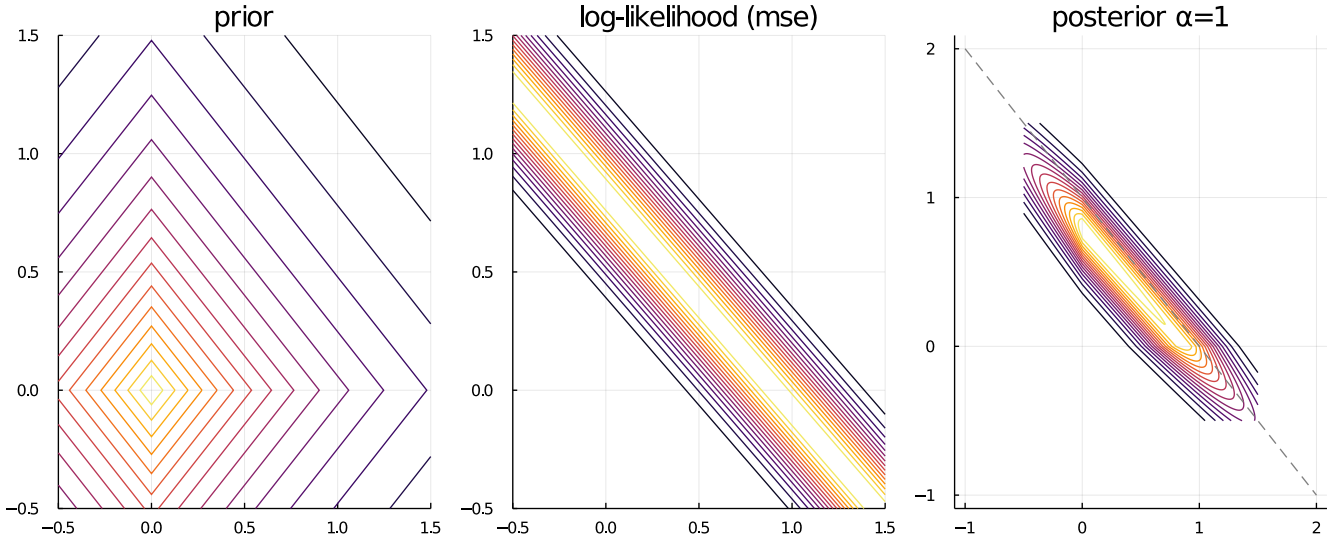
Known as L0 norm.
Penalization the tightest **convex** approximation of L0 norm.



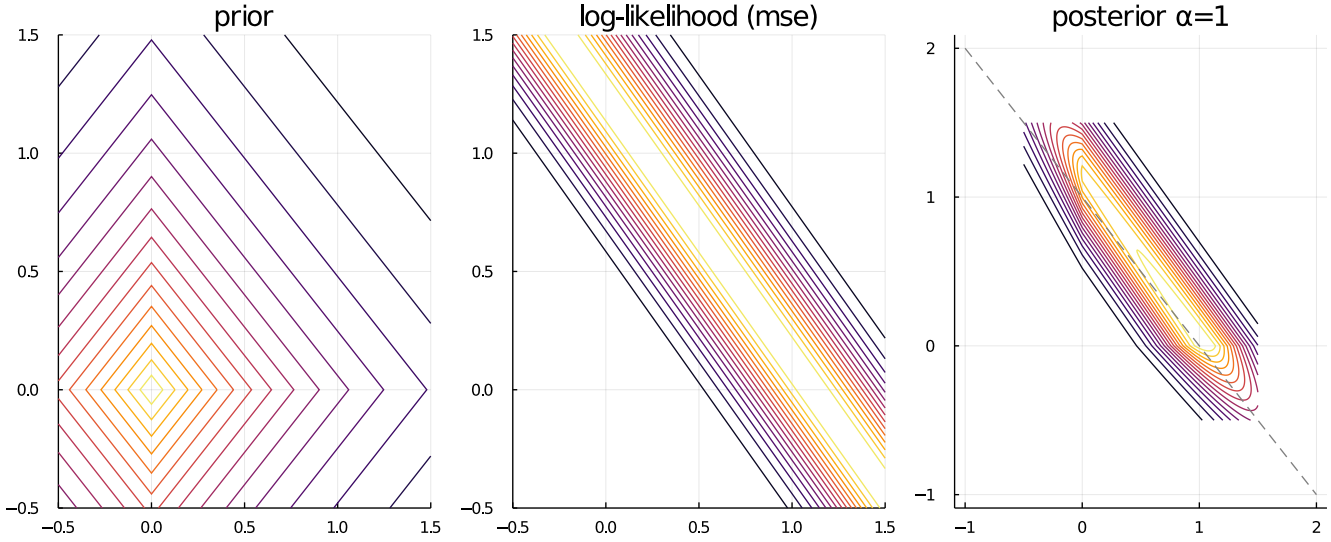
Laplace is the most "spiked" prior with unique extrema!
Lp norm corresponds to Generalized Gaussian.

Uniqueness of Lasso: scaling issue

$x = [0, 1.1], y = [0, 1] \quad \Rightarrow \quad y = x$

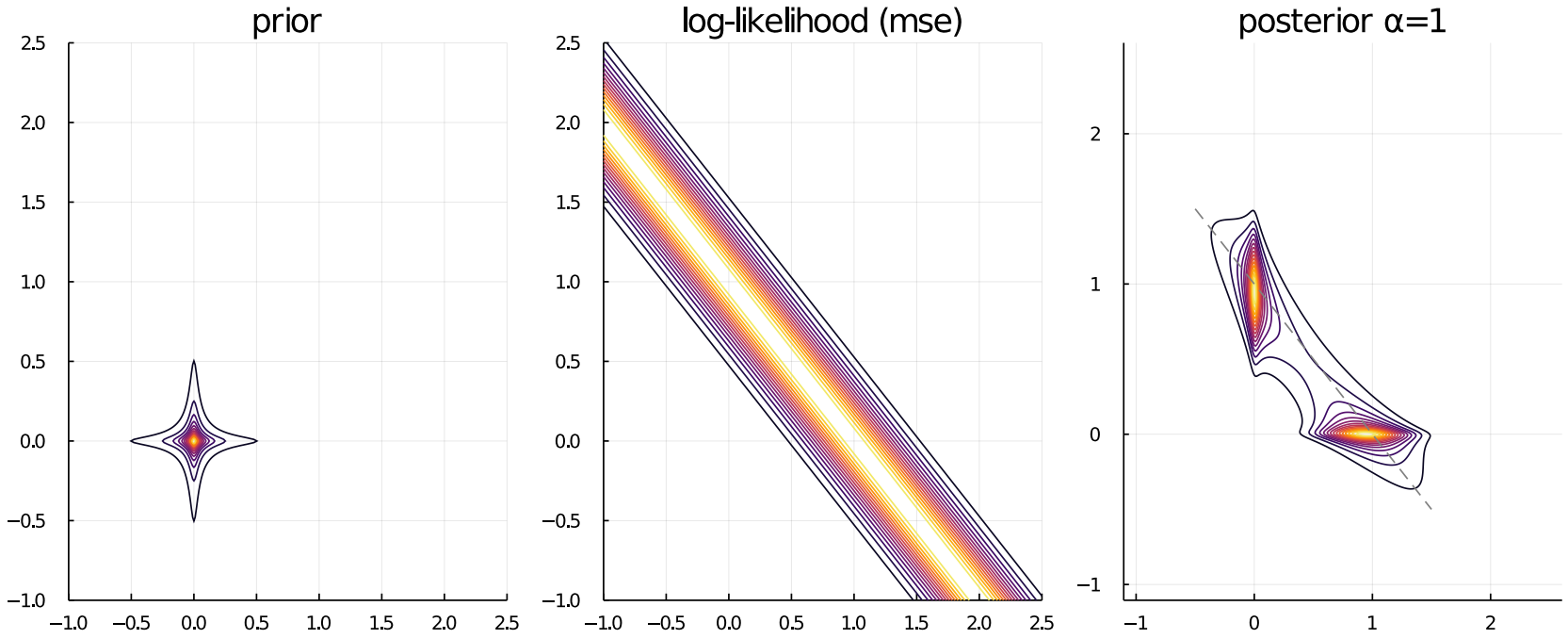


$x = [0, 0.9], y = [0, 1] \quad \Rightarrow \quad y = x^2$



Student't prior, $\mathcal{R} = \log(1+\theta^2/\nu)$

Consider $p(\theta_i) = St(0, \nu, \sigma_0), i = 1, \ldots, p$ i.e. $p(\theta) = \prod_{i=1}^{p} St(0, \nu, \sigma_0)$.
For $\nu = 0.001, \sigma_0 = 1$:



Multiple solutions. Scaling affects which one is global.

- How to do Bayesian inference?
- Student't is not conjugate.

The power of latent variable

Student't is a marginal of Gauss with inverse Gamma prior

$$p(\theta|s) = \mathcal{N}(\mu, s), \qquad p(s) = i\mathcal{G}(\gamma, \delta),$$
$$p(\theta|s) = \mathcal{N}(\mu, \omega^{-1}), \qquad p(\omega) = \mathcal{G}(\gamma, \delta),$$

yields:

$$p(\theta) = \int p(\theta|s)p(s)ds = St(2\gamma, \mu, (\gamma\delta)^{-0.5}).$$

Motivates to replace the Bayes rule:

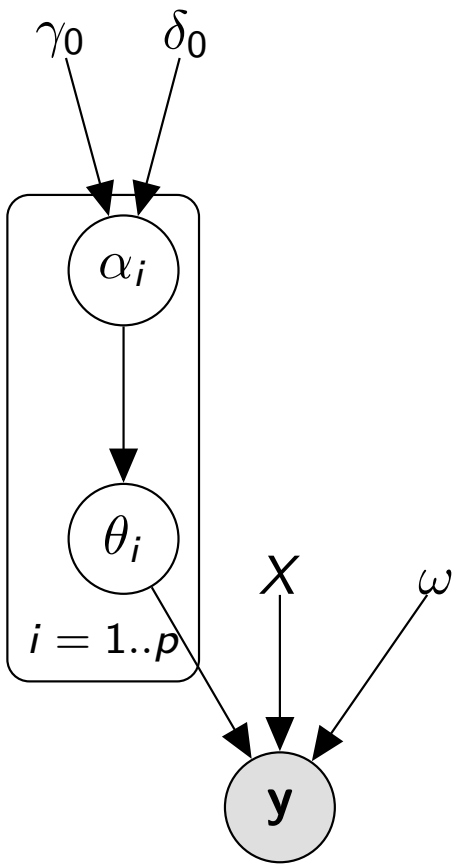
$$\begin{aligned} p(\theta|X, \boldsymbol{y}) &\propto p(\boldsymbol{y}|X, \theta) \int p(\theta|s)p(s)ds \\ &\propto \int p(\boldsymbol{y}|X, \theta)p(\theta|s)p(s)ds \\ &= \int p(\theta, s|X, \boldsymbol{y})ds \end{aligned}$$

Solving $p(\theta, s|X, \boldsymbol{y})$ e.g. by Variational Bayes $p \approx q(\theta)q(s)$.

Automatic relevance determination (ARD)

Probability model

$$
\begin{aligned}
p(\mathbf{y}, \theta | X, \alpha, \omega) &= p(\mathbf{y}|\theta, X, \omega) p(\theta|\alpha) \\
&= \mathcal{N}(X\theta, \omega I)\mathcal{N}(0, \mathrm{diag}[\alpha_1, \ldots, \alpha_p]^{-1}) \\
&\propto \exp\left\{ -\frac{1}{2}\omega ||\mathbf{y} - X\theta||_2^2 - \frac{1}{2}\sum_i \alpha_i \theta_i^2 \right\}
\end{aligned}
$$



Introduce prior

$$
p(\alpha_i) = G(\delta_0, \gamma_0), \quad p(\boldsymbol{\alpha}) = \prod_i p(\alpha_i)
$$

compute

$$
p(\boldsymbol{\alpha}|\mathbf{y}, X)
$$

or

$$
p(\theta|X, \mathbf{y}) = \int p(\theta, \alpha|X, \mathbf{y}) d\alpha
$$

.

Variational Bayes for Automatic relevance determination

Probability model

$$p(\mathbf{y}, \theta | X, \alpha, \omega) = \mathcal{N}(X\theta, \omega I)\mathcal{N}(0, \mathrm{diag}[\alpha_1, \ldots, \alpha_p]) \prod_i G(\gamma_0, \delta_0)$$

Posterior factors

$$\begin{aligned}
p(\alpha_i | \mathbf{y}, X) &= G(\gamma, \delta_i), \\
&\qquad \delta_i = \delta_0 + \frac{1}{2}\mathrm{E}\left(\theta_i^2\right), \ \gamma = \gamma_0 + \frac{1}{2}, \\
p(\theta | \mathbf{y}, X) &= \mathcal{N}(\hat{\theta}, \Sigma_\theta), \\
&\qquad \hat{\theta} = (\omega X^T X + \mathrm{diag}\mathrm{E}\left(\alpha\right))^{-1} \omega X^T \mathbf{y}, \\
&\qquad \Sigma_\theta = (\omega X^T X + \mathrm{diag}\mathrm{E}\left(\alpha\right))^{-1}. \\
p(\omega | \boldsymbol{y}, X) &= G(\delta_0 + p/2, \mathrm{E}((\boldsymbol{y} - X\theta)^T (\boldsymbol{y} - X\theta)))
\end{aligned}$$

Iterated least squares. Moment

Variational Bayes for Automatic relevance determination

Probability model

$$p(\mathbf{y}, \theta | X, \alpha, \omega) = \mathcal{N}(X\theta, \omega I)\mathcal{N}(0, \mathrm{diag}[\alpha_1, \ldots, \alpha_p]) \prod_i G(\gamma_0, \delta_0)$$

Posterior factors

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p(\alpha_i | \mathbf{y}, X) &= G(\gamma, \delta_i), \\
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p(\theta | \mathbf{y}, X) &= \mathcal{N}(\hat{\theta}, \Sigma_\theta), \\
&\qquad \hat{\theta} = (\omega X^T X + \mathrm{diag}\mathrm{E}\left(\alpha\right))^{-1} \omega X^T \mathbf{y}, \\
&\qquad \Sigma_\theta = (\omega X^T X + \mathrm{diag}\mathrm{E}\left(\alpha\right))^{-1}. \\
p(\omega | \boldsymbol{y}, X) &= G(\delta_0 + p/2, \mathrm{E}((\boldsymbol{y} - X\theta)^T (\boldsymbol{y} - X\theta)))
\end{aligned}$$

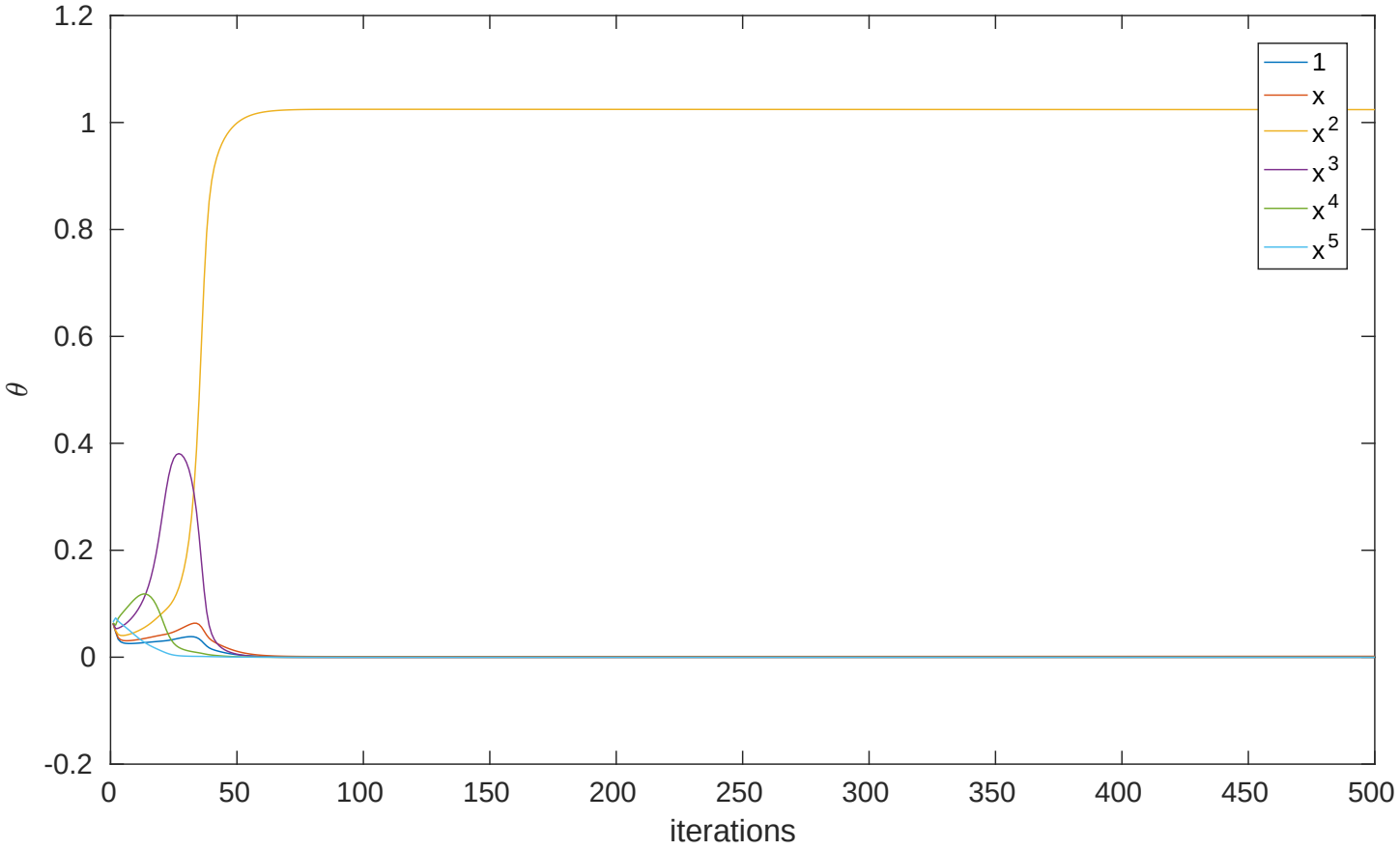
Iterated least squares. Moment

$$\mathrm{E}\left(\theta_i^2\right) = \hat{\theta}_i^2 + \Sigma_{\theta,i,i}, \quad \mathrm{E}(\alpha_i) = \frac{\gamma}{\delta_i}, \quad \mathrm{E}(\theta^T M \theta) = \hat{\theta}^T M \hat{\theta} + \mathrm{tr}(M\Sigma_\theta)$$

Variational Bayes for Automatic relevance determination

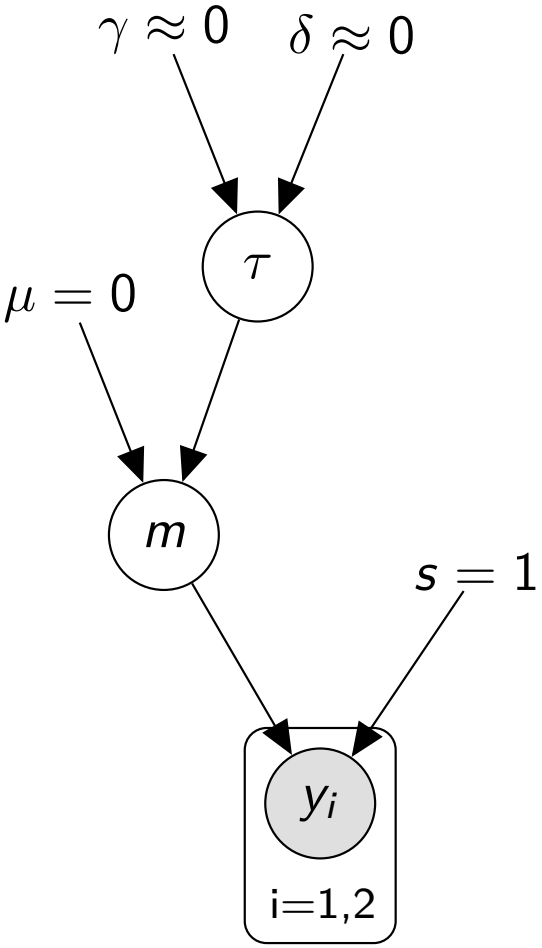
Convergence for

$$y = x\theta_1 + x^2\theta_2 + x^3\theta_3 + x^4\theta_4 + x^5\theta_5$$



Posterior approximation is a Gaussian. => single mode. Local.

Revisiting toy example:

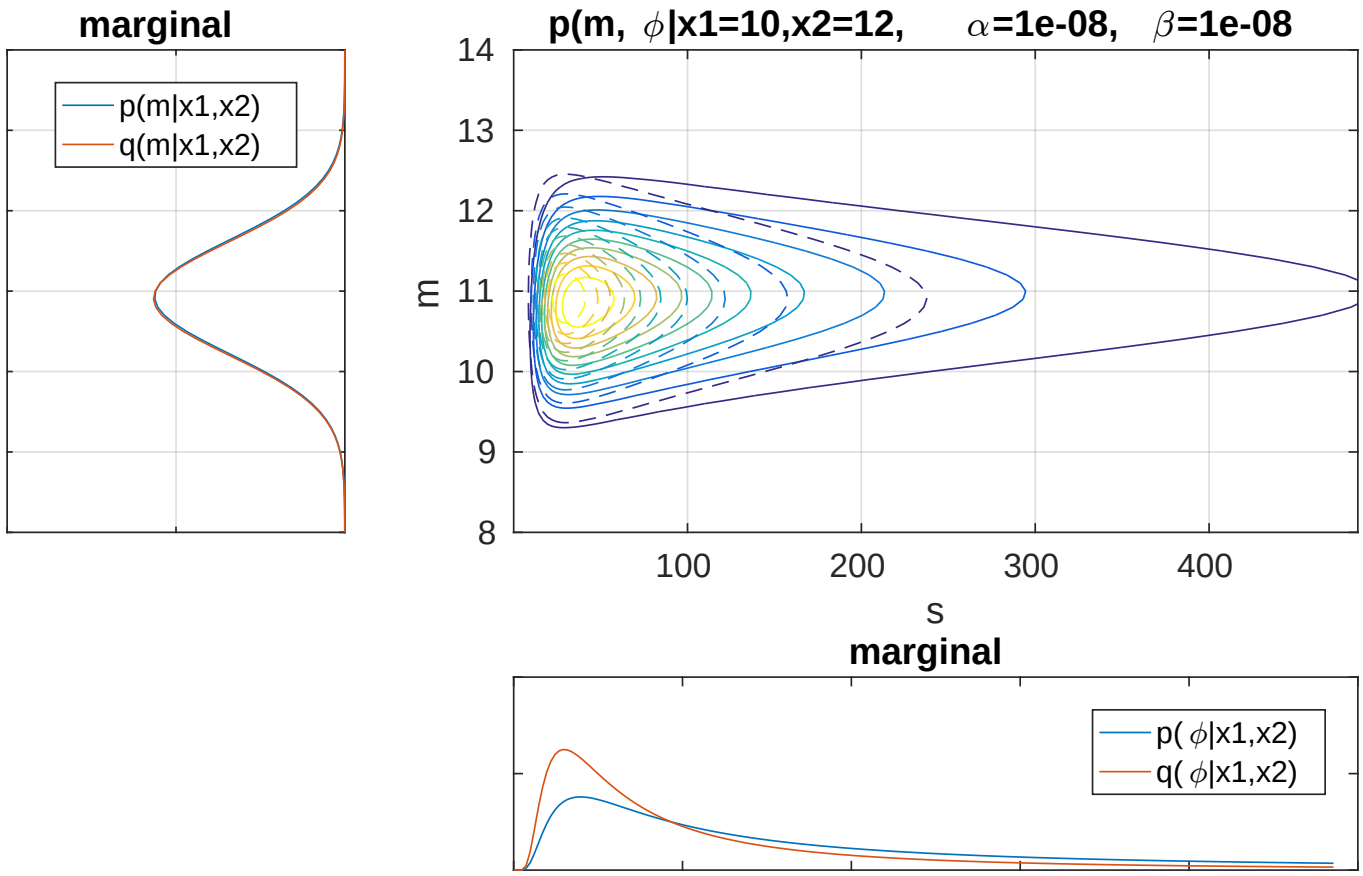


$$
\begin{aligned}
p(y_i|m) &= \mathcal{N}(m, 1), \\
p(m|\tau) &= \mathcal{N}(0, \tau), \\
p(\tau) &= iG(\gamma_0, \delta_0)
\end{aligned}
$$

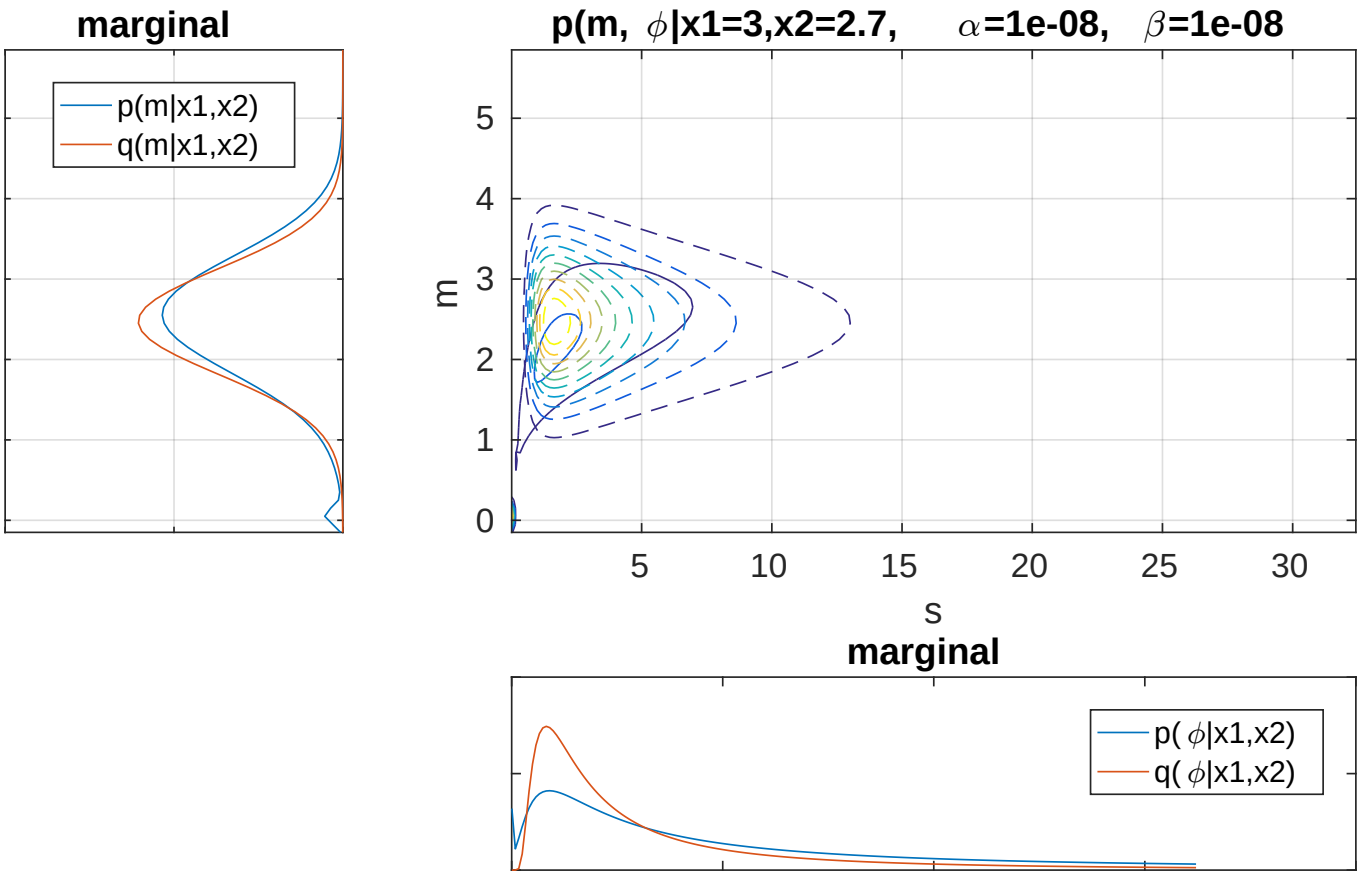
Solution using:

- Variational Bayes
- Numerical evaluation on grid

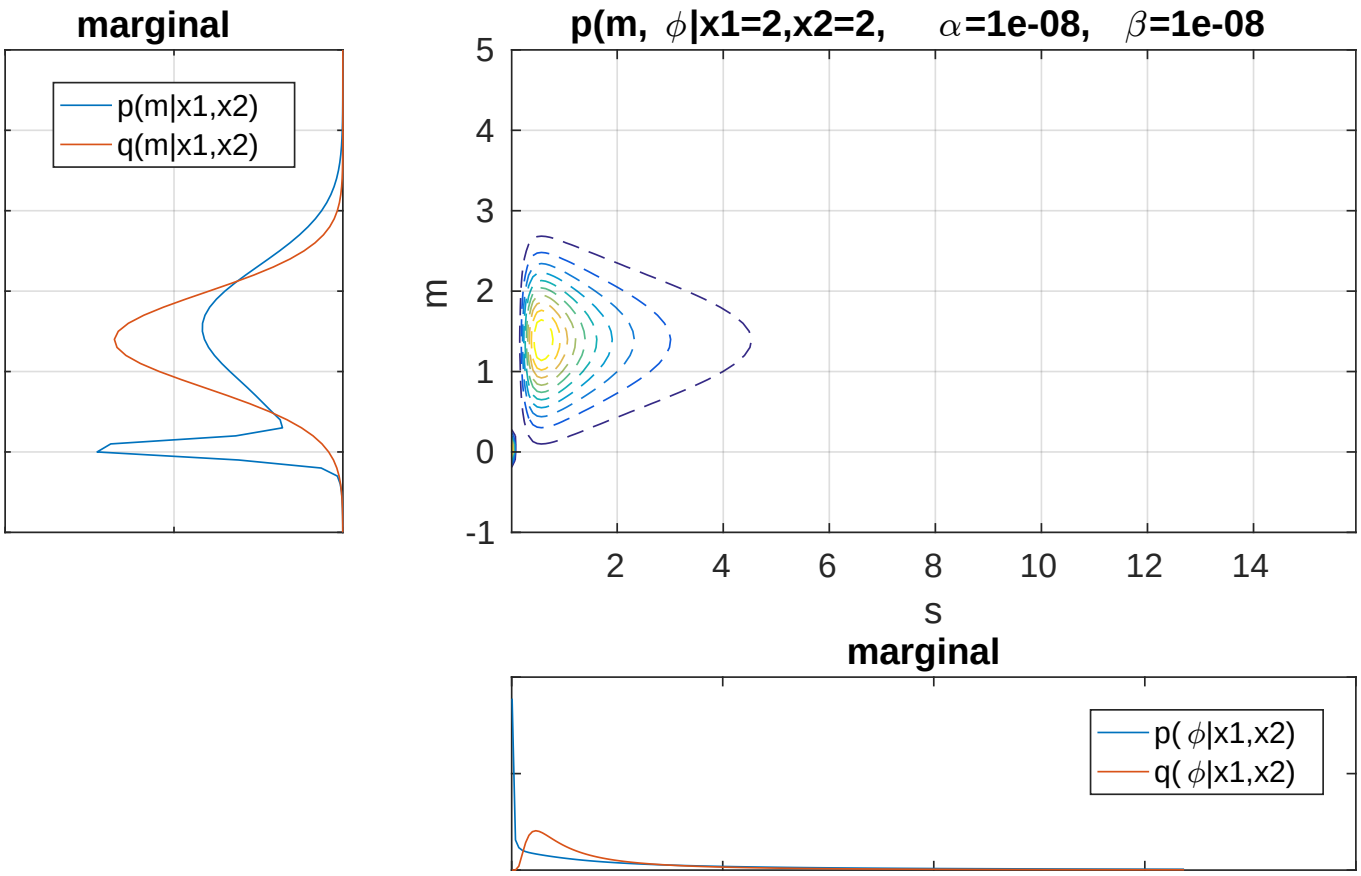
Revisiting toy example:



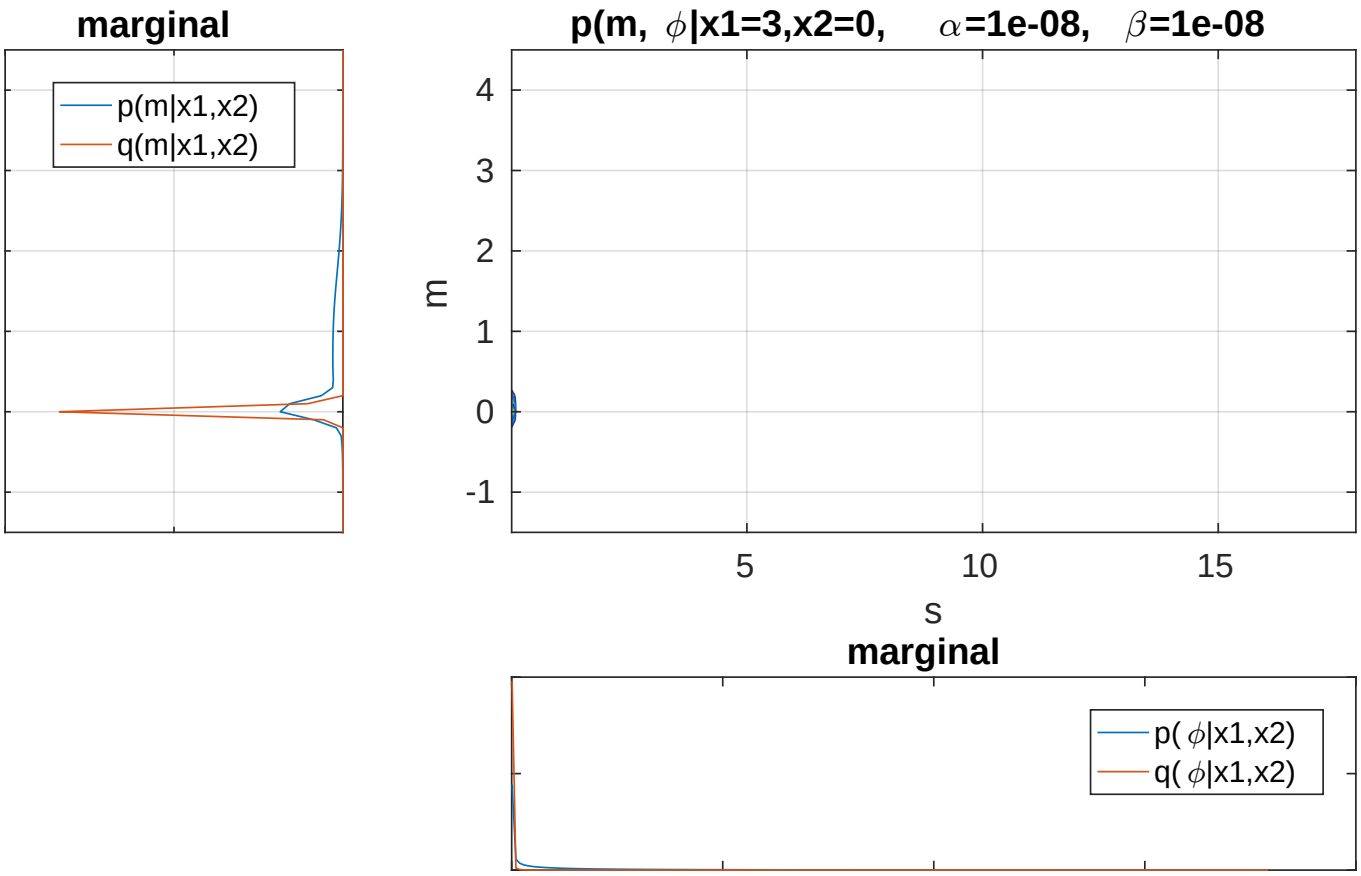
Revisiting toy example:



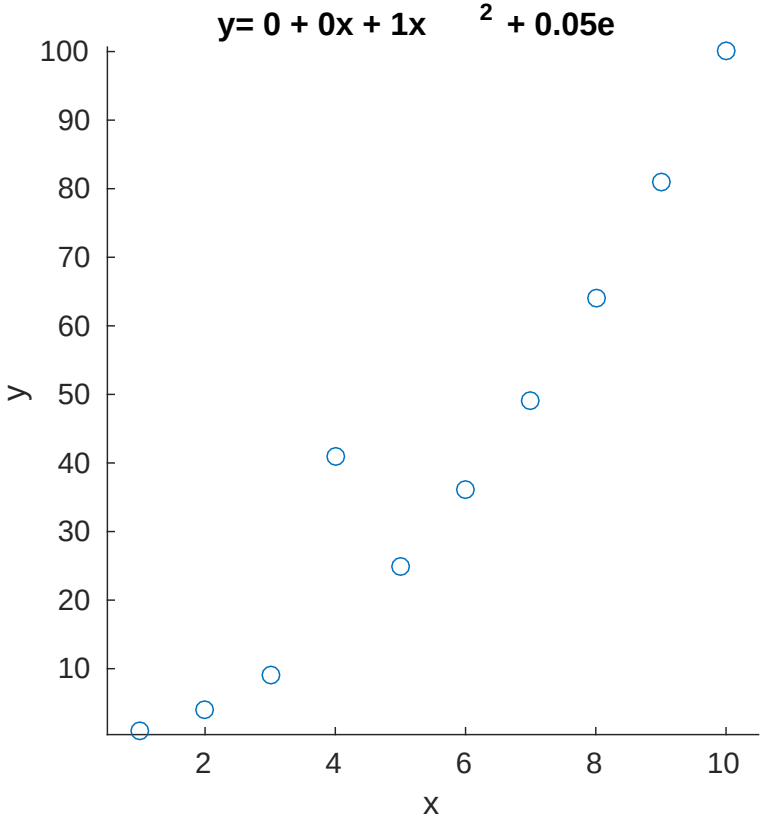
Revisiting toy example:

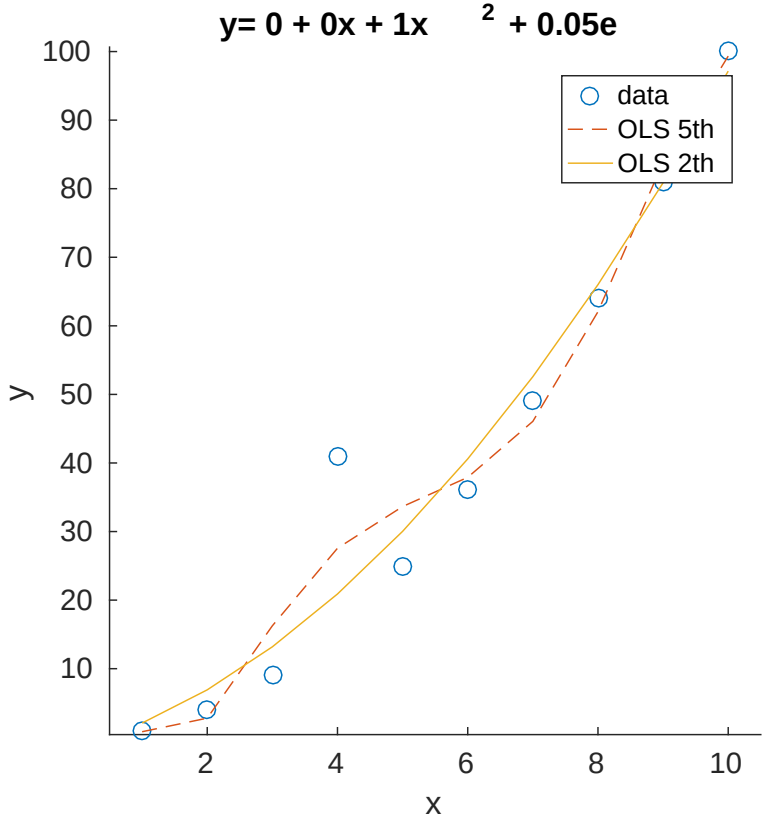


Revisiting toy example:



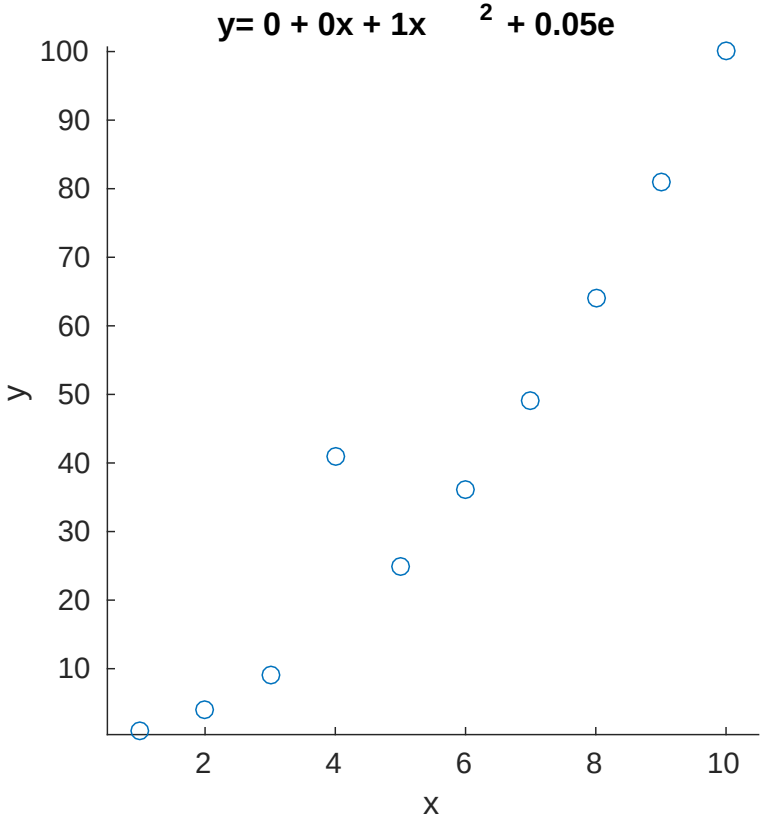
Outliers

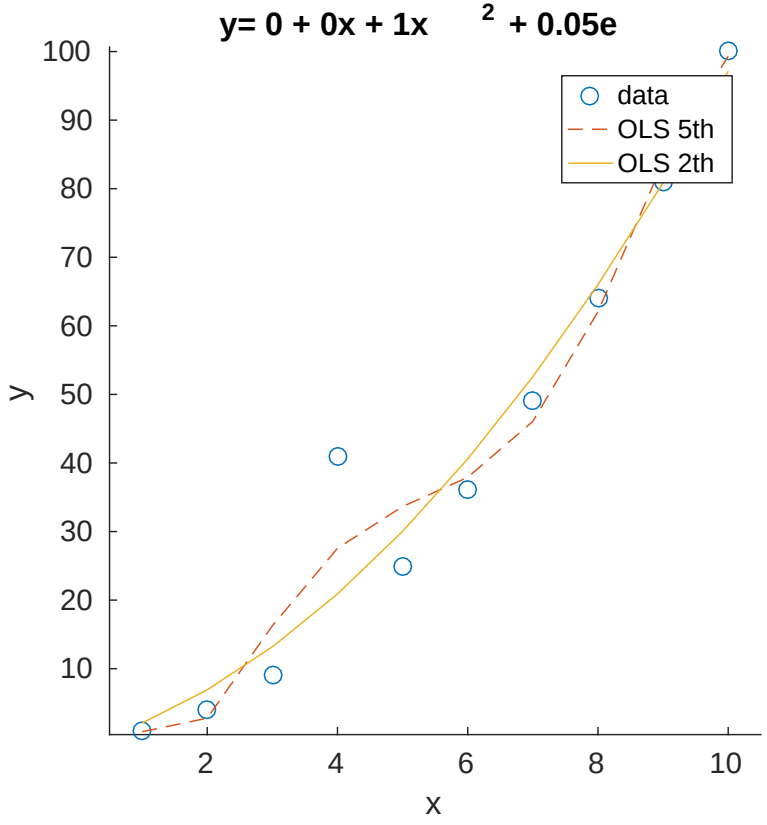




- How to minimize the effect of an outlier?

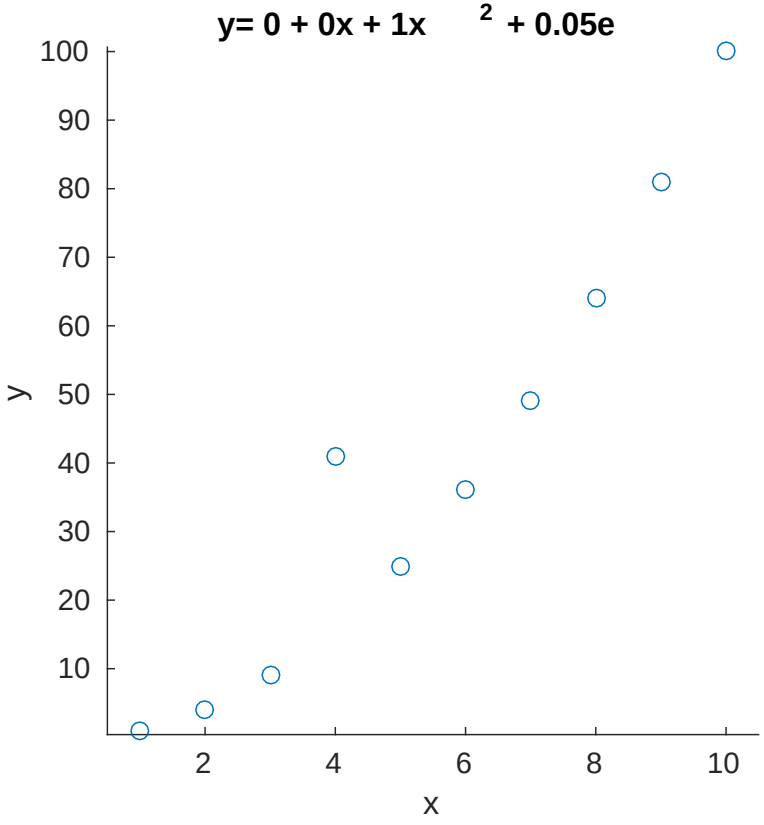
Outliers

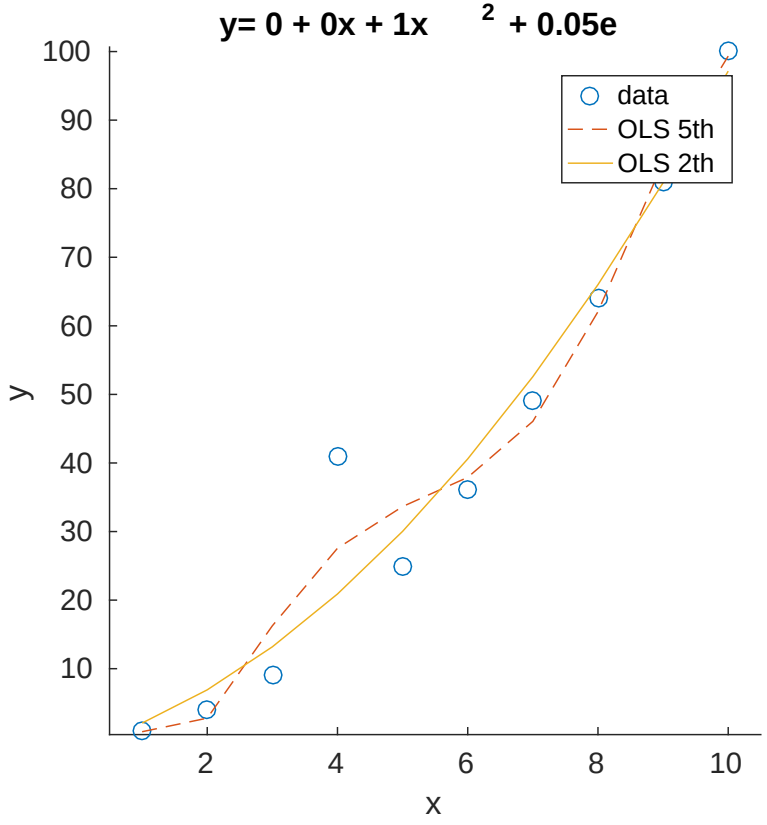




- How to minimize the effect of an outlier?
- Outlier detection, robust statistics, etc.

Outliers





- How to minimize the effect of an outlier?
- Outlier detection, robust statistics, etc.
- Hierarchical model?

Outliers hierarchical model

Probability model

$$
\begin{aligned}
p(\mathbf{y}, \theta | X, \alpha) &= p(\mathbf{y} | \theta, X) p(\theta | \alpha) \\
&= \mathcal{N}(X\theta, \beta^{-1} I) \mathcal{N}(0, \alpha^{-1} I).
\end{aligned}
$$

Outliers hierarchical model

Probability model

$$\begin{aligned} p(\mathbf{y}, \theta | X, \alpha) &= p(\mathbf{y} | \theta, X) p(\theta | \alpha) \\ &= \mathcal{N}(X\theta, \beta^{-1} I) \mathcal{N}(0, \alpha^{-1} I). \end{aligned}$$

Is the variance of the noise homogenous?

Outliers hierarchical model

Probability model

$$\begin{aligned} p(\mathbf{y}, \theta | X, \alpha) &= p(\mathbf{y} | \theta, X) p(\theta | \alpha) \\ &= \mathcal{N}(X\theta, \beta^{-1} I) \mathcal{N}(0, \alpha^{-1} I). \end{aligned}$$

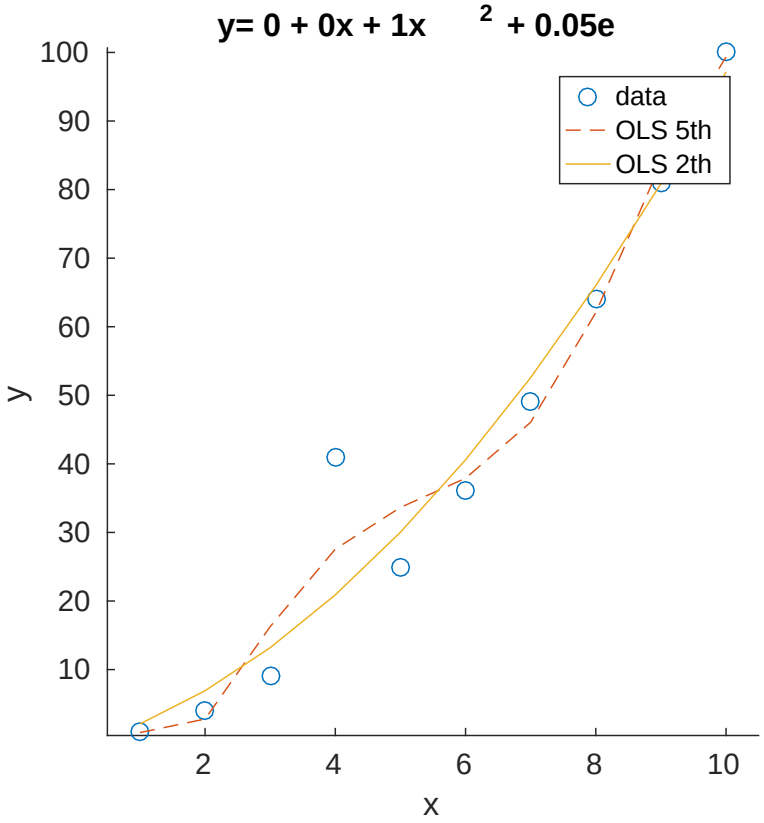
Is the variance of the noise homogenous?
New model:

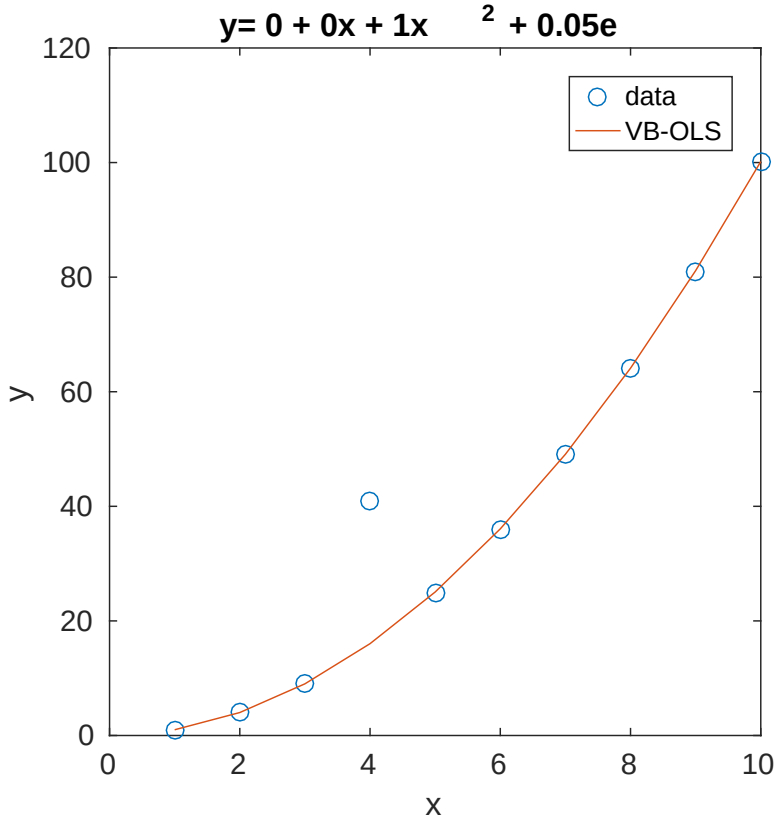
$$\begin{aligned} p(\mathbf{y}, \theta | X, \alpha) &= p(\mathbf{y} | \theta, X) p(\theta | \alpha) \\ &= \mathcal{N}(X\theta, \mathrm{diag}[\beta_1, \ldots, \beta_n]) \mathcal{N}(0, \alpha^{-1} I). \end{aligned}$$

Prior

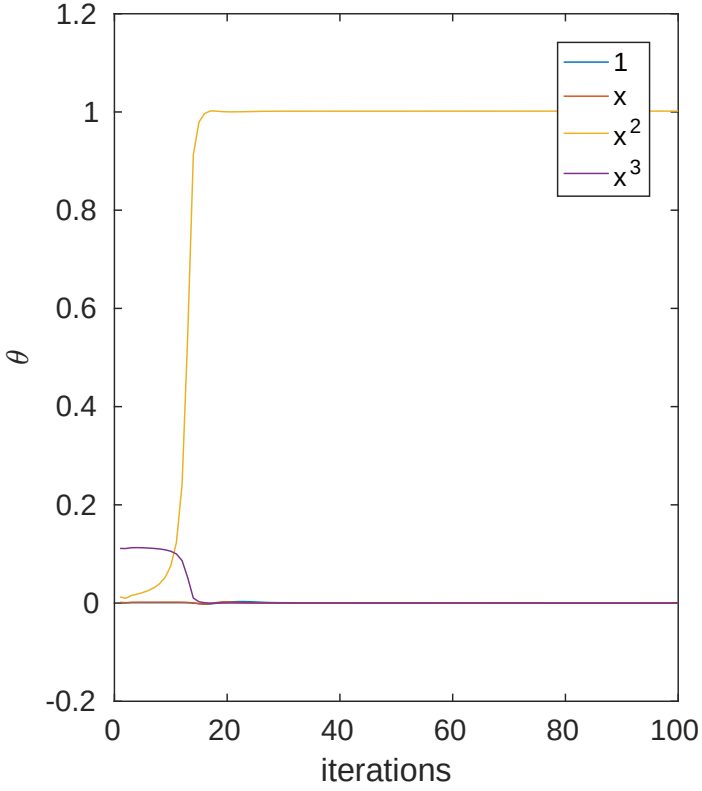
$$p(\beta_i) = G(\delta, \gamma), \;\; p(\boldsymbol{\beta}) = \prod_i p(\beta_i)$$

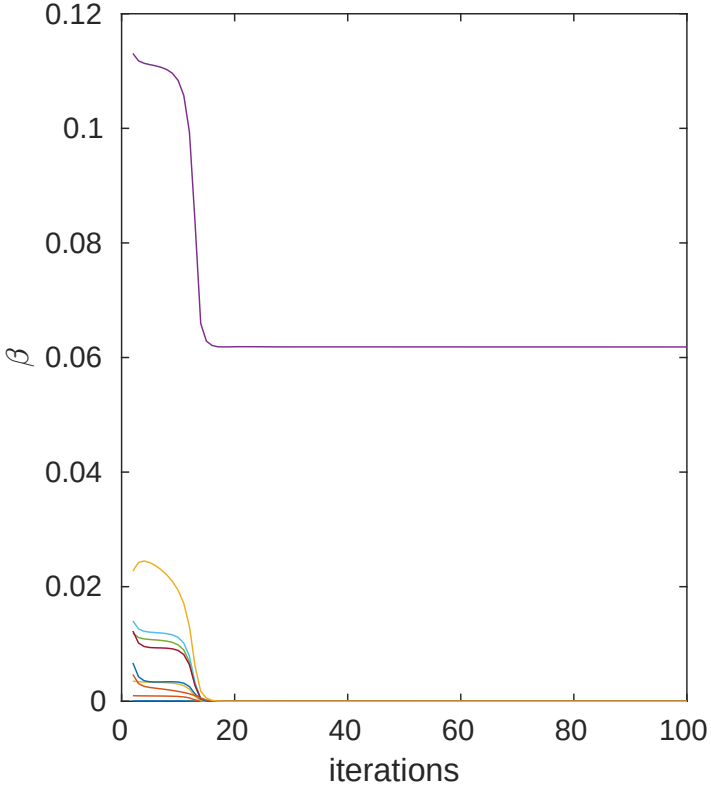
Outliers


y= 0 + 0x + 1x 2 + 0.05e
data
OLS 5th
OLS 2th
y
x
100
90
80
70
60
50
40
30
20
10
2
4
6
8
10


y= 0 + 0x + 1x 2 + 0.05e
data
VB-OLS
y
x
120
100
80
60
40
20
0
0
2
4
6
8
10

Outliers, both diagonal α and β





- local minima, unstable for both α and β.

Conclusion

- Linear regression is solved by OLS.
- When the data are not informative, we need to regularize:
- Different prior assumptions yield different results
  - ridge regression minimizes coefficients
  - sparsity prior minimizes the number of non-zero coefficients
- Non-Gaussian residues
  - Student-t residue,
  - Mixture residue, etc.

ETEX data challenge

Estimate source term of the ETEX experiment

$$y = X\theta,$$

where θ is assumed to be sparse, piece-wise linear, with non-negative elements.

	points
least squares estimate θ	5
using "non-convex" prior (e.g. ARD)	5

Beware:

- scale of the matrix is very low! Normalize to $X/\max(X)$